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NATURAL DISASTER EARLY PREDICTION AND RESPONSE SYSTEM

Monitoring, Alerting and Emergency Communication Platform

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1. INTRODUCTION

Recent advances in embedded systems, wireless communication, and mobile computing have created new opportunities for developing integrated, real-time environmental monitoring solutions. However, many current approaches remain fragmented, with sensing, data processing, and user interaction often treated as separate components rather than a unified system.

This project proposes an integrated approach that combines distributed embedded devices with a mobile application, enabling seamless data flow from physical sensors to end users. The system is designed to be modular and scalable, allowing the integration of various sensor types and supporting deployment across different environments and use cases.

Although the underlying architecture is adaptable to multiple hazards, this work focuses specifically on landslides. Despite their significant impact, landslides are frequently underrepresented in technological solutions, particularly in comparison to other natural disasters. By addressing this domain, the project demonstrates how localized and often overlooked risks can be effectively monitored using accessible technology.

At the same time, the proposed solution is conceived as a foundation for a broader multi-hazard platform, with the potential to expand toward monitoring other natural phenomena such as floods, earthquakes, or wildfires.

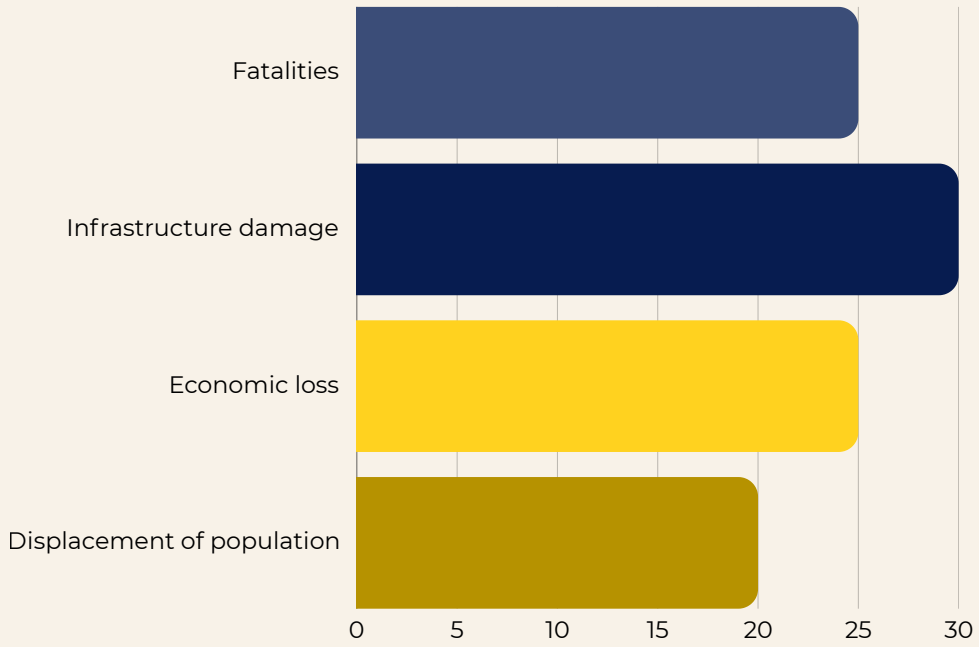
2. LANDSLIDES: CHARACTERISTICS AND IMPACT

Landslides are a type of geological hazard that involve the movement of soil, rock, or debris down a slope due to gravity. They are typically triggered by a combination of environmental factors, including prolonged rainfall, rapid snowmelt, earthquakes, volcanic activity, and human interventions such as deforestation or construction.

One of the primary causes of landslides is increased soil moisture, which reduces the cohesion between particles and decreases slope stability. Additional indicators include ground deformation (tilt changes), micro-vibrations, and surface cracks, all of which can signal an increased probability of slope failure.

Landslides occur worldwide and represent a significant risk to both human life and infrastructure. According to global disaster databases, landslides account for thousands of fatalities annually and cause substantial economic damage, particularly in mountainous and rural regions where monitoring systems are limited or nonexistent.

Certain regions are more vulnerable due to terrain and climate conditions. Areas with steep slopes, loose soil composition, and high precipitation levels are especially prone to landslide events. In addition, climate change is contributing to an increased frequency of extreme weather conditions, further elevating the risk.



Impact of Landslides (Global Perspective)

3. DEFINING THE PROBLEM

Natural disasters such as landslides, extreme weather conditions, ground instability and vibrations pose a serious threat to the safety of people and infrastructure, especially in areas where there are no continuous monitoring systems. In many environments, especially rural and geographically complex, early warning systems are unavailable, too expensive or not adapted to local conditions, which is why dangerous situations are often detected too late.

Existing solutions mainly rely on centralized infrastructure and expensive sensor networks, which limits their application and speed of reaction at the local level. Also, there is often a lack of direct communication with users, so information about risk, even when gathered, does not reach those who need it most in a timely manner.

Therefore, there is a need for a compact, energy-efficient and affordable system that can continuously monitor environmental conditions, recognize early signs of danger and warn users in a timely manner, with the possibility of direct communication with rescue services.

4. PROPOSED SOLUTION

To address the identified challenges, this project proposes an integrated system that combines distributed embedded sensor nodes with a user-oriented mobile application. The system is designed to continuously monitor environmental conditions, analyze potential risk factors, and provide timely alerts to users and responders.

At the hardware level, sensor units equipped with moisture, tilt, and vibration sensors collect real-time data from landslide-prone areas. This data is transmitted through a wireless communication layer to a cloud-based infrastructure, where it is processed and transformed into a unified risk score.

At the software level, the mobile application serves as the main interface between the system and its users. It visualizes environmental data, displays risk levels, and enables immediate response through an integrated SOS feature. Additionally, it supports user awareness and preparedness through educational content and interactive modules.

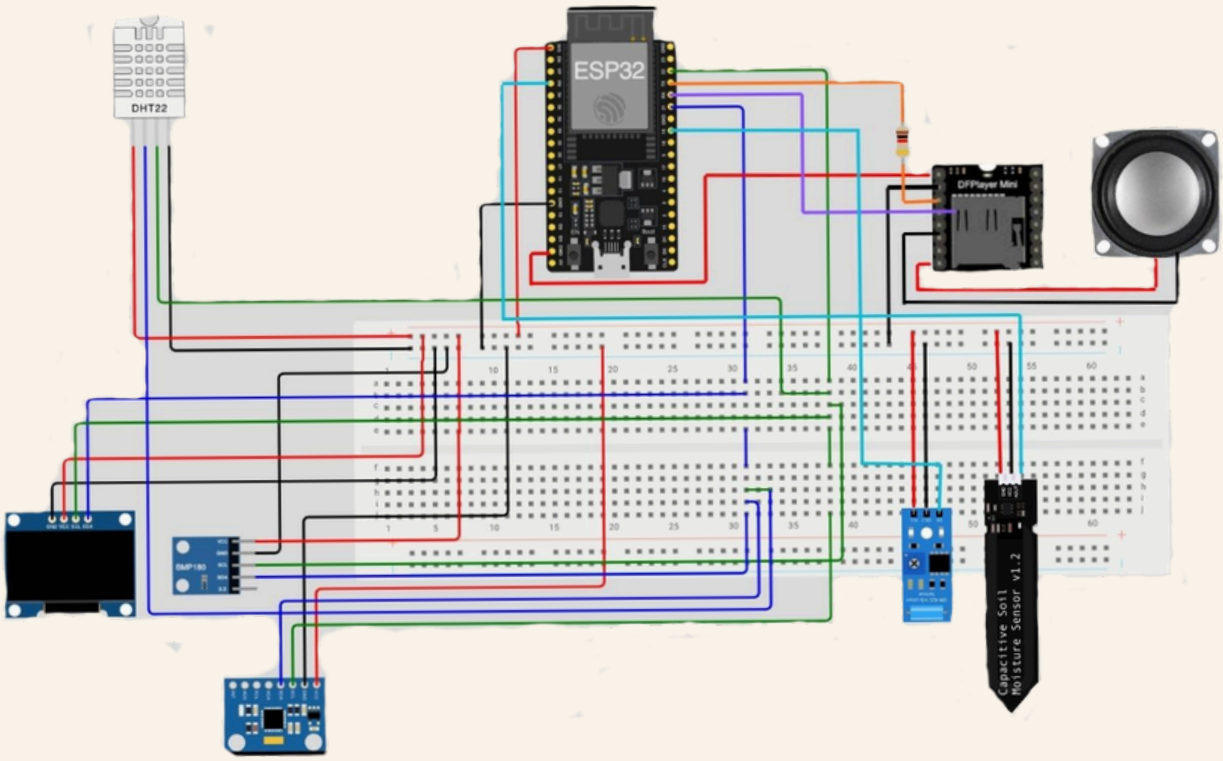
The system is designed with scalability and adaptability in mind, allowing integration of additional sensors and expansion to other types of natural hazards. By combining real-time monitoring, data-driven analysis, and direct user interaction, the proposed solution aims to provide a practical and effective approach to early warning and disaster risk reduction.

5. ELECTRONICS DESIGN

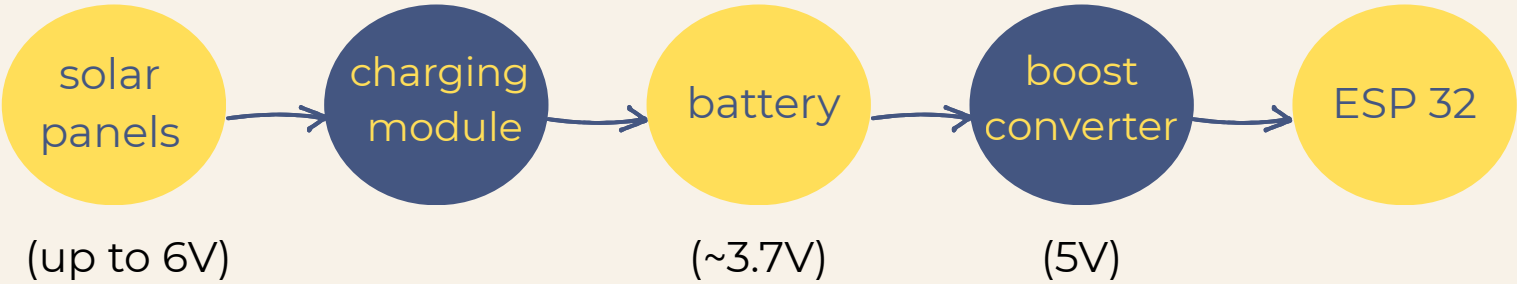
The electronic design of the system is based on the ESP32 microcontroller, which is the central control unit responsible for collecting, processing and distributing data. The ESP32 was chosen for its high processing power, low power consumption and support for various communication protocols, which makes it suitable for integrating multiple sensors into a single system.

The BMP280, MPU6050 sensors and the OLED display are connected via I2C communication using the SDA and SCL lines, which allows for efficient data exchange with a minimum number of pins. This approach simplifies the design and allows for parallel connection of multiple devices to the same bus. The DHT22 temperature and humidity sensor is connected to a digital pin, while the soil moisture sensor is connected to an analog input, which allows for accurate reading of continuous values. The vibration sensor uses a digital input to detect sudden changes in the environment.

Based on the system diagram, the signal flow goes from the sensors to the ESP32, where the data is processed and then sent to output components such as LED indicators, OLED display and DFPlayer. Power for all components comes from a common source via a boost converter, ensuring stable system operation. This architecture allows for modularity, easy expansion and reliable communication between all elements of the system.

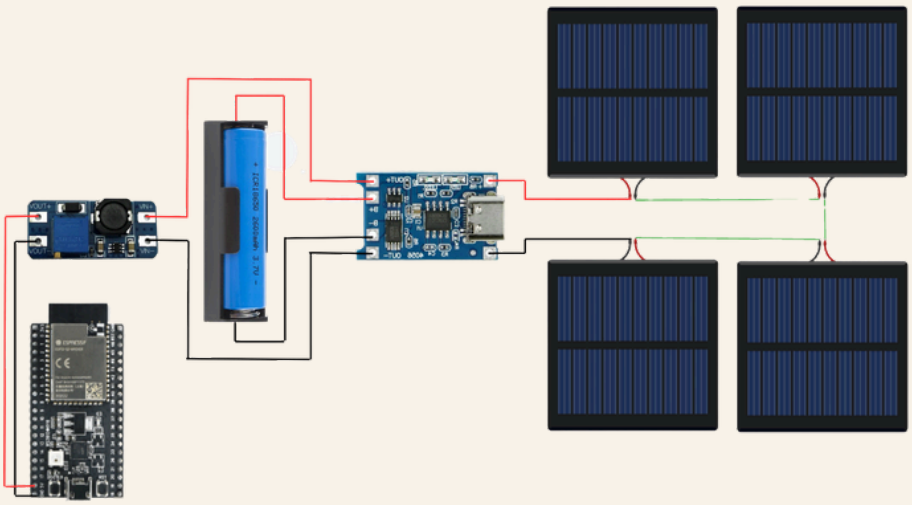


5.1 POWER MANAGEMENT SYSTEM



The system is powered by a 18650 Li-Ion battery that enables long-term operation in autonomous conditions, and energy is restored via a solar panel. A TP4056 module is used for charging management, which ensures safe battery charging with protection against overcharging and overdischarging, thereby extending the battery life. The battery output voltage (3.0–4.2V) is not sufficient for stable operation of all system components, so a boost converter was implemented that raises the voltage to a stable 5V. This allows reliable power supply to the ESP32 microcontroller and all peripheral devices even when the battery is partially discharged.

Additionally, consumption optimization is achieved using the ESP32 deep sleep mode, where the system adjusts the sleep mode depending on the current risk level. This strategy significantly reduces average energy consumption and enables long-term autonomous operation of the system, which is especially important for remote and difficult-to-reach areas where access to the electrical network is limited or non-existent.



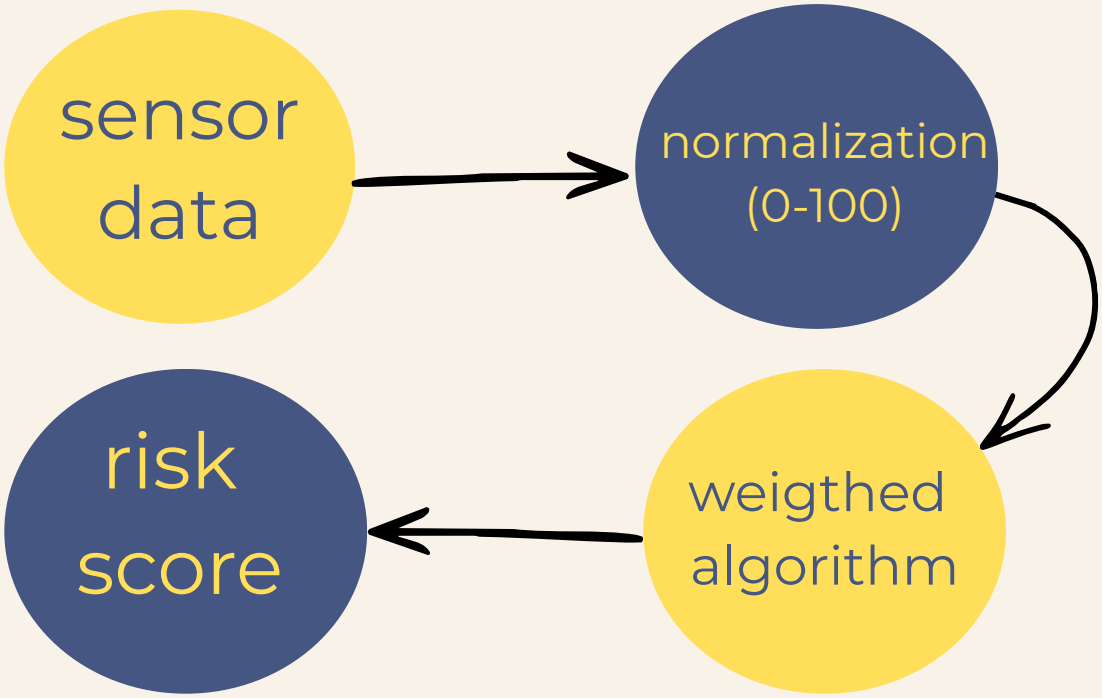
5.2 SENSOR DATA PROCESSING AND RISK ANALYSIS

The system collects data from multiple sensors that measure various physical parameters of the environment, then converts them into standardized factors ranging from 0–100 for easy comparison.

Soil moisture is mapped to a factor that indicates potential soil instability, vibration is detected as a binary signal and directly increases risk, while acceleration with the MPU6050 detects sudden movements or changes in structure. Weather conditions are estimated based on temperature and atmospheric pressure.

The total risk score is calculated as a weighted sum:
risk = (soil × 0.3) + (vibration × 0.25) + (acceleration × 0.25) + (weather × 0.2)

The weights reflect the relative importance of each factor, with soil having the largest impact and weather conditions adding to it. The result is a percentage of risk and enables simple and efficient real-time data analysis, suitable for embedded systems with limited resources.



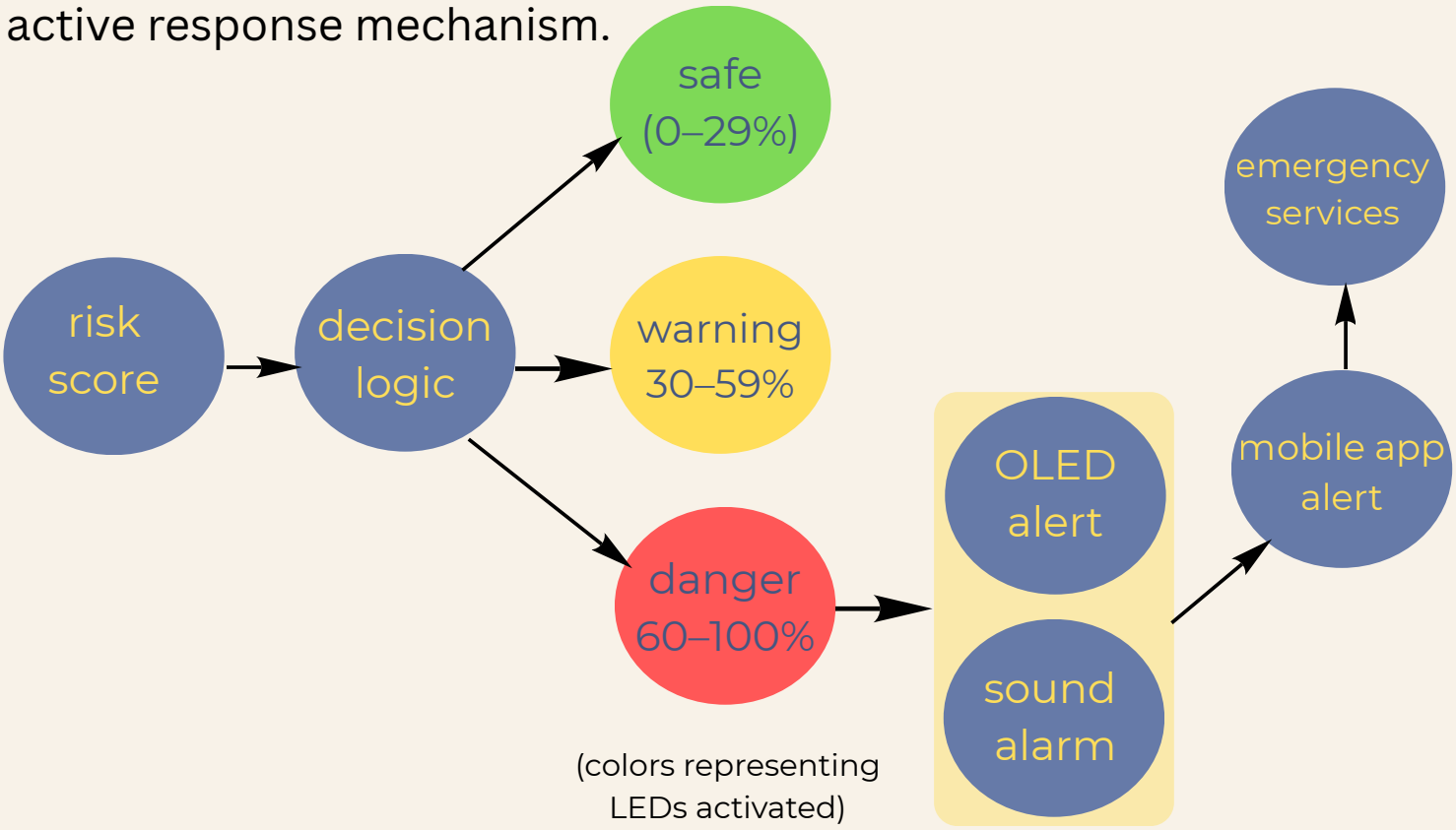
5.3 ALERT AND RESPONSE SYSTEM

The warning system operates by monitoring the calculated risk score and determining the system state: Safe, Warning or Danger.

In Safe mode, the system continues normal monitoring, Warning signals increased risk and the need for caution, while Danger activates all warning mechanisms for a timely response.

LED indicators visually display the risk level, the OLED display shows the current score and status, and the DFPlayer mini plays an audio alarm in critical situations.

The system is connected to a mobile application that sends notifications and enables remote monitoring. In case of serious danger, the application enables direct communication with rescue services, which switches the system from passive monitoring to an active response mechanism.

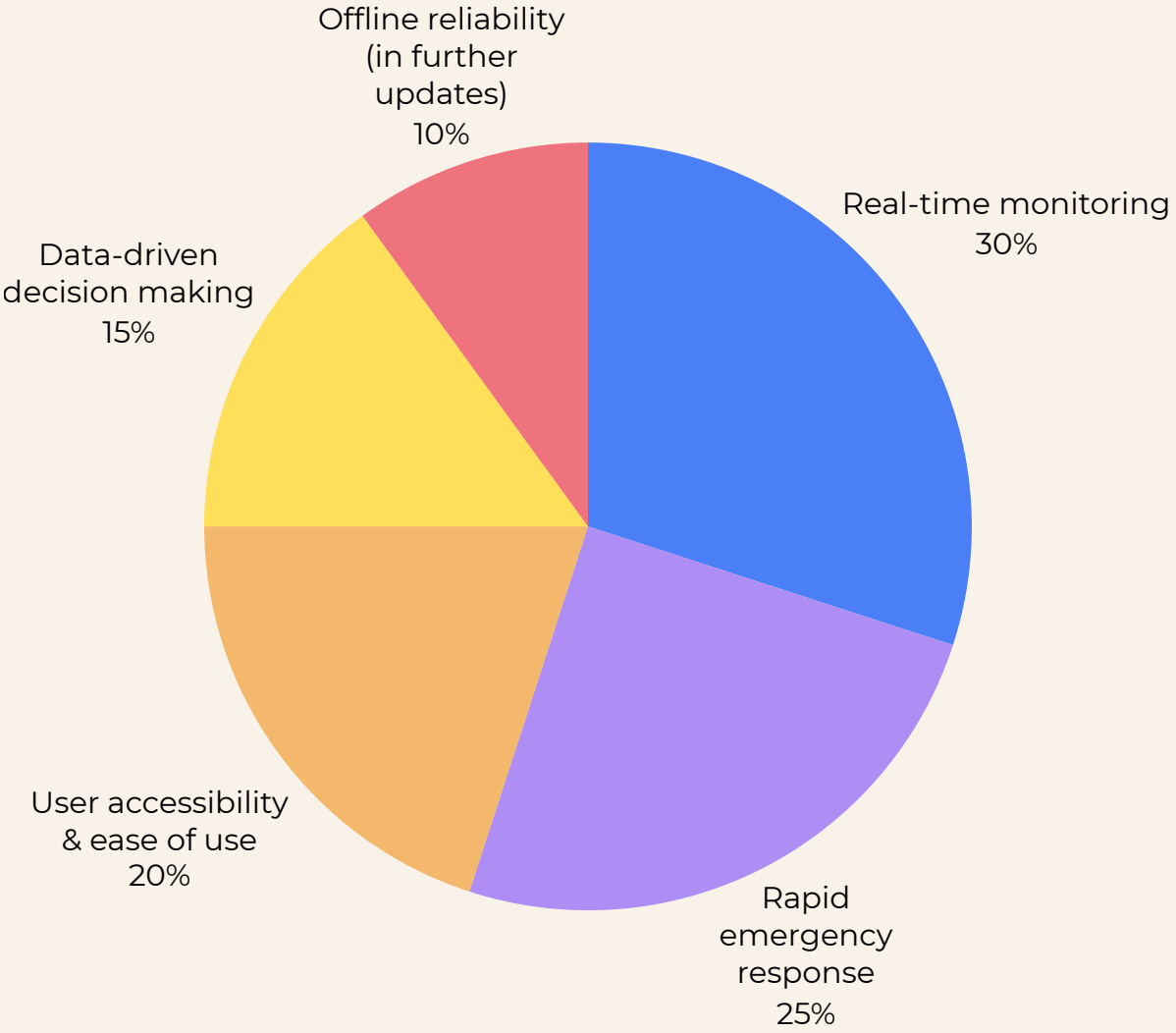


6. MOBILE APPLICATION DESIGN & FUNCTIONALITY

6.1 Overview

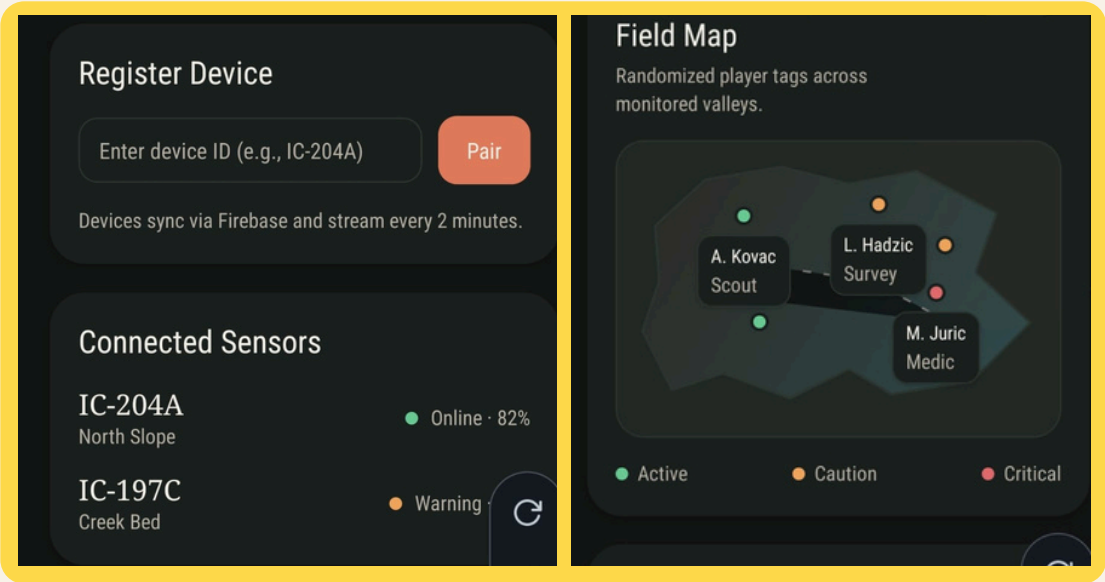
The mobile application serves as the central user interface of the system, enabling real-time monitoring, early warning visualization, and emergency response coordination. It integrates data collected from distributed sensor nodes and translates it into actionable insights for users, communities, and first responders.

The application is structured into four main modules: Early Warning, SOS Dispatch, Education & Preparedness, and Device Management.



6.2 DEVICE CONNECTIVITY

The mobile application enables seamless integration with field-deployed sensor devices through a cloud-based synchronization system. Each device is uniquely identified and connected via a secure communication channel, allowing continuous transmission of environmental data.
Once registered, devices periodically update their status, including connectivity, battery level, and sensor readings. This ensures that users have real-time insight into system performance and can quickly detect any malfunction or anomaly in the sensor network.



6.3 FIELD MAP VISUALIZATION

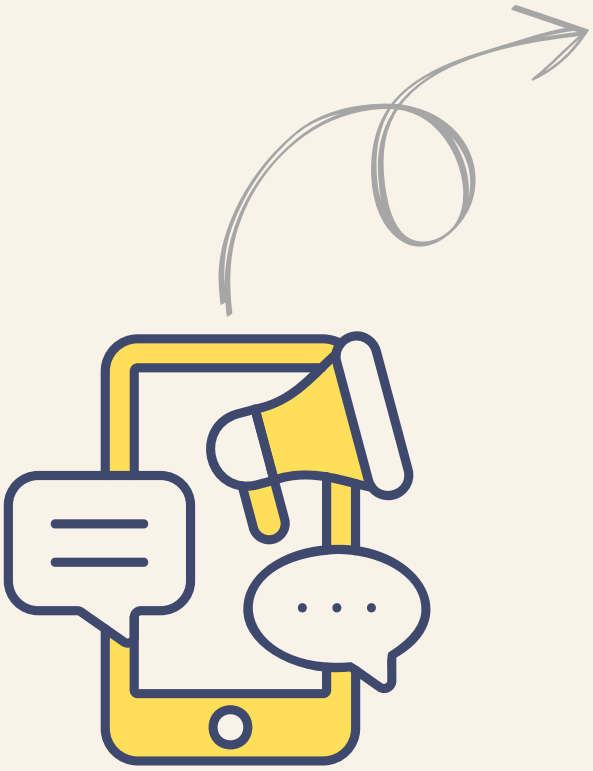
The application features an interactive field map that provides a real-time spatial overview of monitored areas. Sensor nodes and active users are displayed using color-coded indicators representing different risk levels, such as safe, caution, and critical zones.
This visualization allows users to quickly identify high-risk regions and understand the geographical distribution of alerts. It also enhances coordination by enabling responders to track locations and prioritize interventions based on real-time conditions.

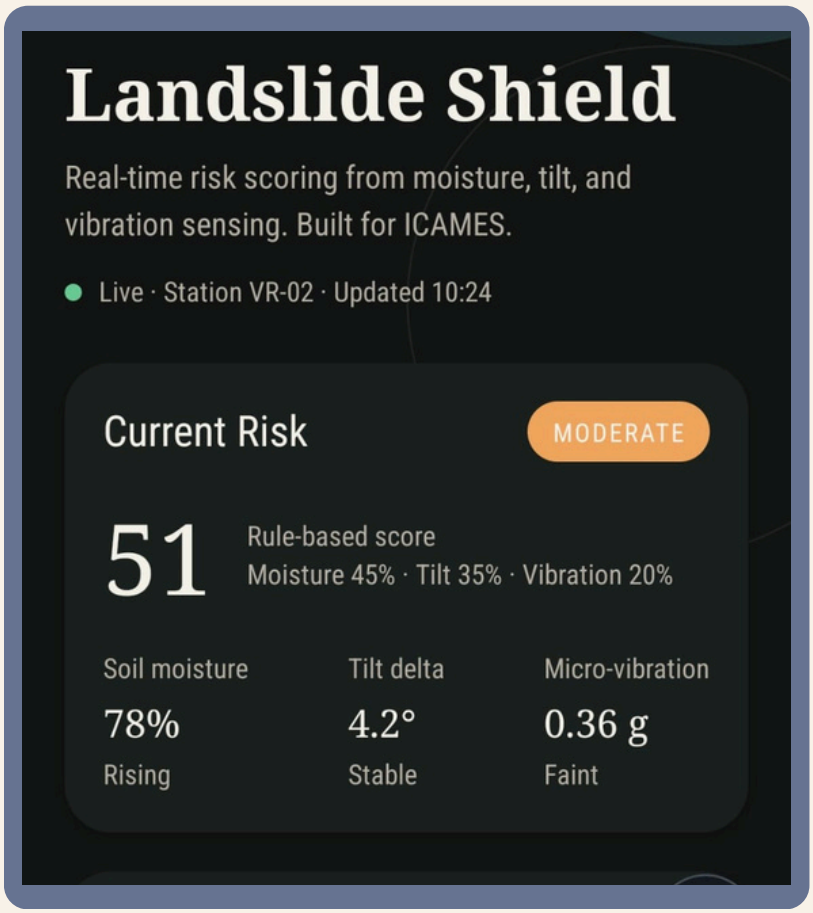
6.4 EARLY WARNING MODULE

The Early Warning interface provides users with real-time situational awareness based on environmental sensor data. It displays a dynamically calculated risk score derived from multiple parameters including soil moisture, tilt variation, and micro-vibrations. These inputs are weighted to produce a unified risk index, allowing users to quickly assess the severity of conditions.

Additionally, the application visualizes signal trends over time, highlighting changes in environmental behavior across the last 24 hours. Alert density mapping further identifies high-risk zones by analyzing clustered warning events.

A live field map presents the spatial distribution of monitored areas, including active participants and sensor coverage. This enables both individual users and teams to understand geographic risk distribution in real time.



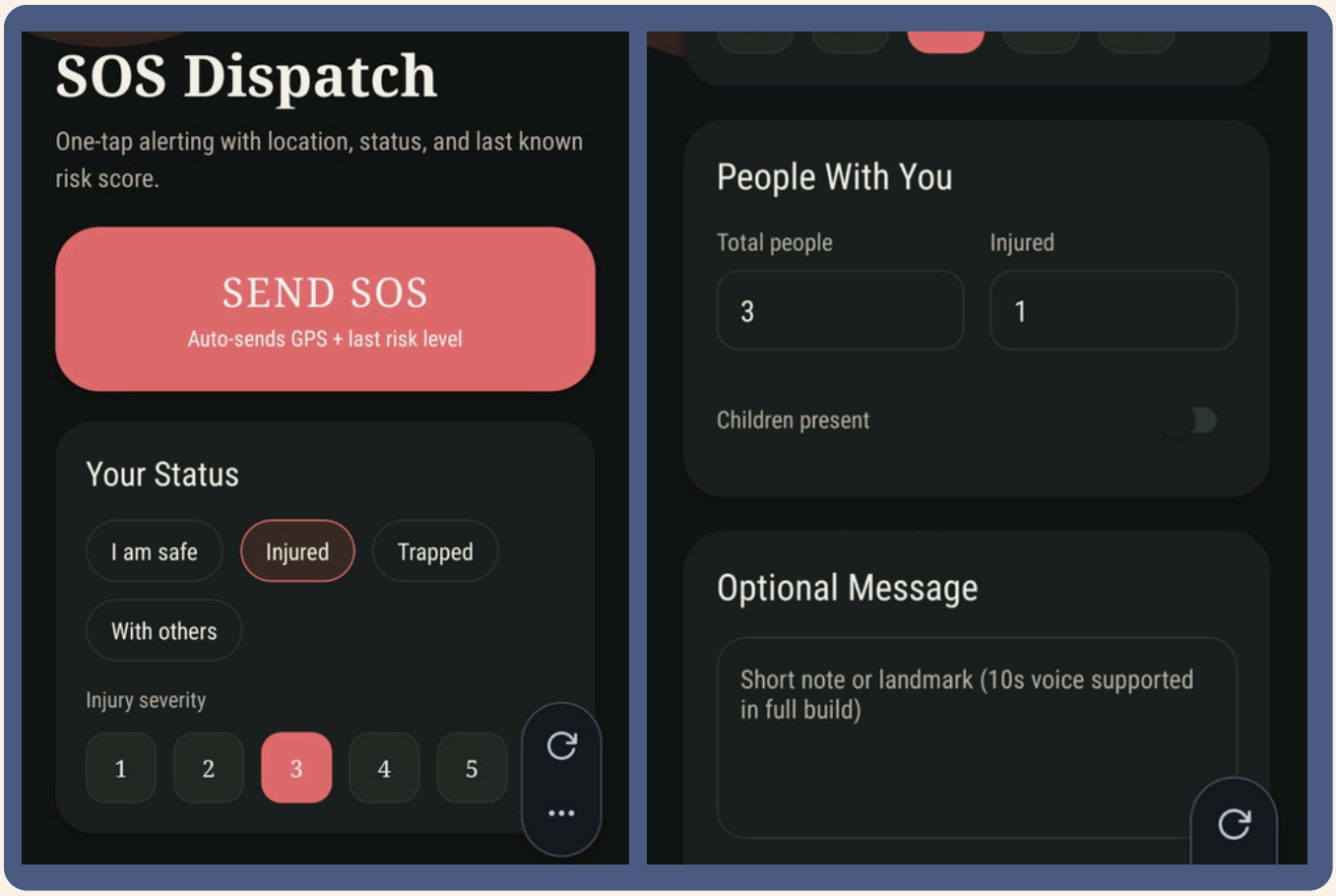


6.5 SOS EMERGENCY DISPATCH SYSTEM

The SOS module is designed for rapid emergency communication with minimal user interaction.
With a single tap, users can send an emergency alert that includes:

1. GPS location
2. current status (safe, injured, trapped)
3. injury severity level
4. number of people involved

This information is transmitted along with the system's latest risk assessment, enabling responders to prioritize interventions effectively.
The system also supports offline queuing, ensuring that alerts are stored and transmitted once connectivity is restored, increasing reliability in disaster scenarios.



6.6 EDUCATION MODULE

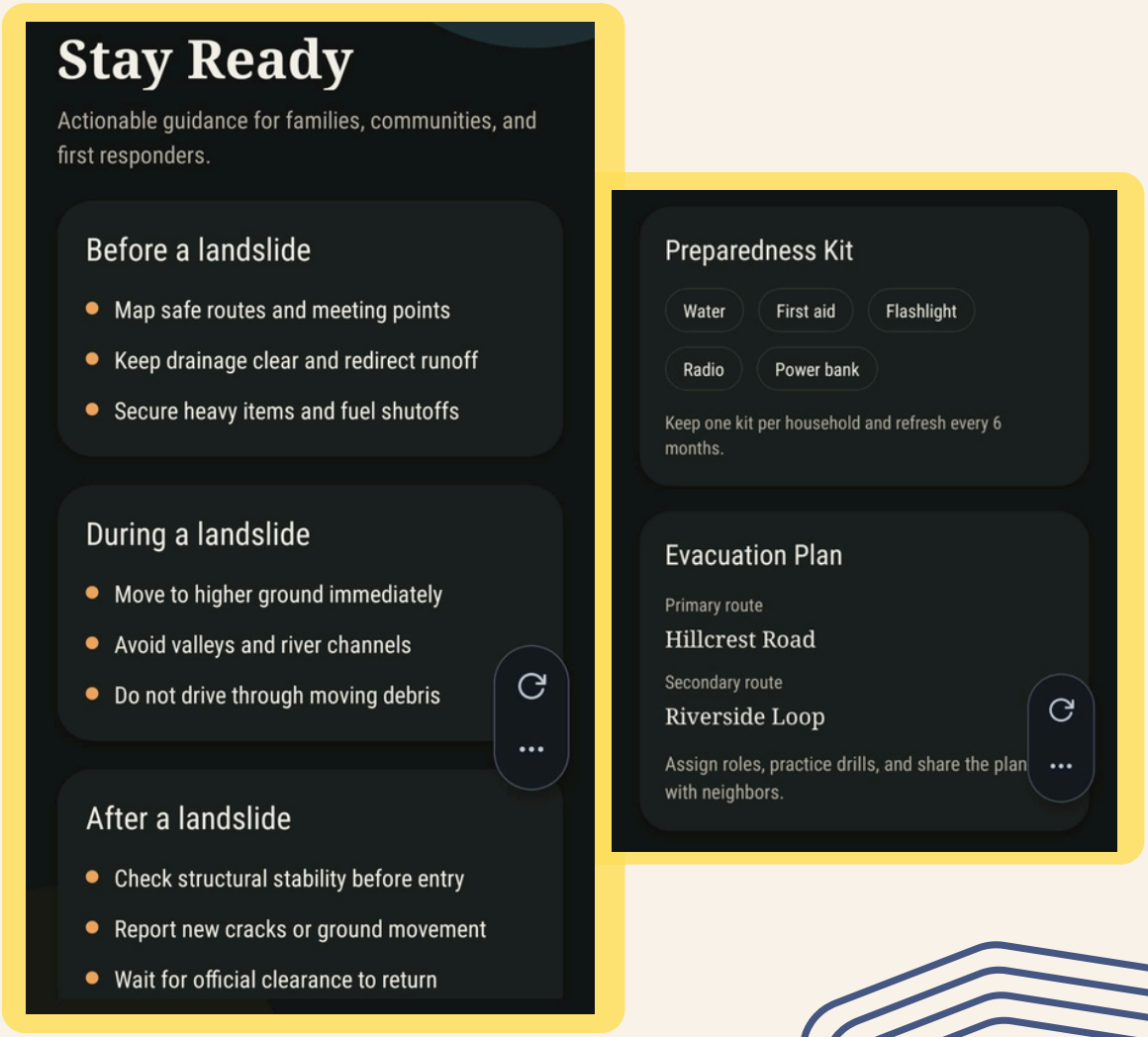
The application includes an integrated educational component aimed at increasing user awareness and preparedness.

It provides structured guidance for:

1. actions before a landslide
2. behavior during an event
3. post-event safety procedures

Additionally, users are guided in assembling emergency preparedness kits and defining evacuation plans, including primary and secondary routes.

This module transforms the application from a passive monitoring tool into an active risk mitigation platform.

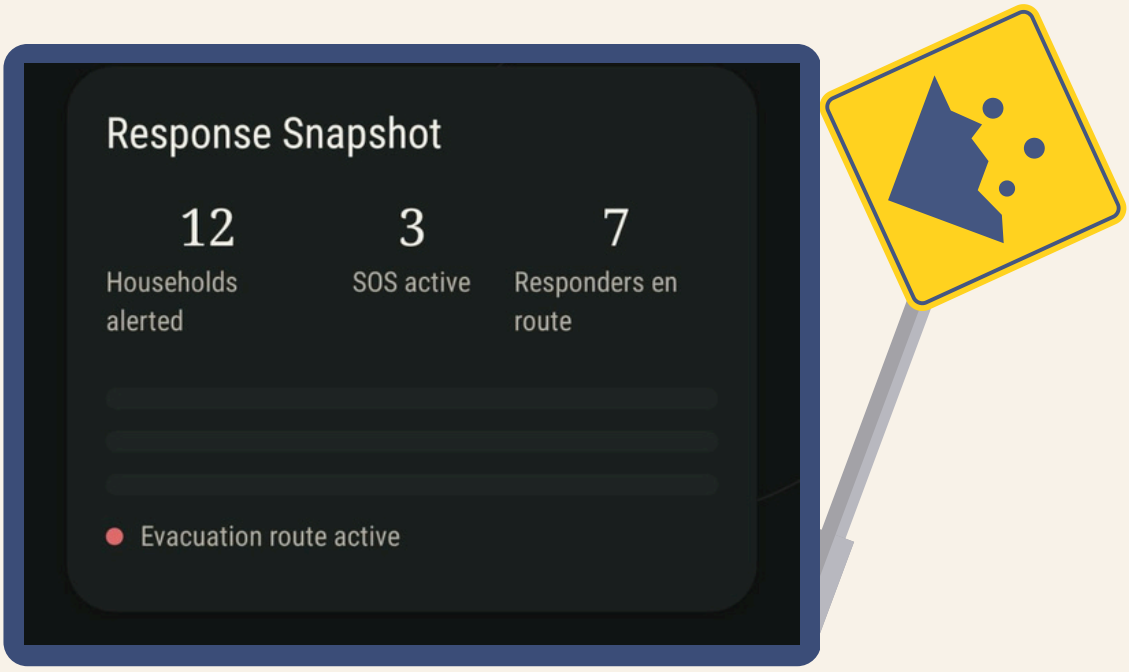


6.7 SYSTEM RESPONSE OVERVIEW

The application also features a response snapshot interface, summarizing ongoing emergency activity.
It includes:

1. number of households alerted
2. active SOS signals
3. responders currently en route

This allows for rapid situational assessment and coordination at both local and system-wide levels.



6.8 DESIGN CONSIDERATIONS

The application is designed with a focus on:

1. minimal user interaction during emergencies
2. clear visual hierarchy for rapid decision-making
3. offline functionality for reliability
4. scalability for multi-user deployment

The interface prioritizes clarity, accessibility, and speed, ensuring usability even under high-stress conditions.

7. ENVIRONMENTAL IMPACT AND COMMUNITY ENGAGEMENT

In addition to technological monitoring, the project also addresses one of the root causes of landslides - soil erosion, which is often intensified by deforestation and uncontrolled land use.
To contribute to long-term risk reduction, the mobile application includes a community-driven initiative aimed at environmental restoration. Through the platform, users will be encouraged to participate in a “Plant a Tree” campaign, promoting reforestation as a natural method for improving soil stability and reducing the likelihood of landslides.
This feature transforms the system from a purely reactive solution into a proactive one, combining real-time monitoring with sustainable prevention strategies. By engaging users directly, the application fosters awareness and collective responsibility, contributing to both environmental protection and disaster risk mitigation.

8. CURRENT PROGRESS AND FUTURE DEVELOPMENT

At the current stage of the project, a functional connection between the embedded system and the mobile application has been successfully established. The embedded system is capable of collecting environmental data through integrated sensors, while the mobile application provides a foundational frontend interface for data visualization and user interaction.
The system architecture, including communication between hardware and software components, has been validated through initial testing, confirming the feasibility of real-time monitoring and data transmission.

Looking ahead, the project is on track to deliver a fully operational prototype by the time of the competition. This includes a complete version of the mobile application capable of real-time data acquisition, processing, and visualization directly from deployed sensor units.
In parallel, the embedded system will be further developed into a fully integrated hardware prototype, incorporating all sensors within a custom-designed 3D-printed enclosure. This will ensure both physical robustness and suitability for real-world deployment in landslide-prone environments.
Together, these developments will result in a comprehensive, end-to-end solution that combines hardware reliability with an intuitive and responsive user interface.

9. CONCLUSION

This project presents an integrated approach to landslide monitoring by combining embedded sensing systems with a user-oriented mobile application. The proposed solution enables real-time data collection, risk assessment, and timely communication, addressing key limitations of existing approaches.

The system has been designed with a strong focus on practicality and real-world applicability, ensuring that it remains both scalable and feasible for deployment in diverse environments. Initial validation of the architecture has been successfully completed, demonstrating the effectiveness of the overall concept.

Based on the current progress, we expect to deliver a fully functional prototype by the time of the competition, including a complete mobile application prototype with real-time data integration, as well as an embedded system with integrated sensors housed in a finalized 3D-designed enclosure.

The proposed solution also provides a foundation for future development into a broader multi-hazard monitoring platform, highlighting its adaptability and long-term potential.

10. MEET OUR TEAM

Ibrahim Čustović, *Project Lead & Embedded Systems Engineer*
Led overall system design and coordination. Developed the ESP32 architecture, sensor integration, and risk analysis algorithm, and contributed to power optimization using deep sleep modes.

Merima Prcanović, *Mobile Application Designer & Presentation Lead*
Developed the mobile app (UI, data visualization, notifications) and handled system communication. Will lead the project presentation and professionally deliver results to the jury.

Davud Sadikaj, *Electronics & Power Systems Engineer*
Responsible for circuit design and hardware implementation, including sensor connections, breadboard setup, and power management system (battery, charging module, boost converter).

Lejla Kukuruzović, *System Integration Engineer*
Handled system integration and testing, ensuring seamless hardware–software operation. Supported communication between components and contributed to debugging and optimization.

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